

Complete System Design Document

Version: 2.0
Date: March 30, 2026
Status: Active Planning — Phase 0 Ready to Build

1. Overview

Dala is a sophisticated workforce management platform designed to replace manual, paper-based processes — specifically the handwritten BDD (Business Development Daily) check-in forms — with a digital, intelligent, and verifiable system.

It combines a smart clock-in engine with layered physical verification, a structured daily productivity check-in (the BDD form), full ERP modules, and an AI intelligence layer — all feeding directly into payroll.

The system is designed around one core principle: **you must prove you are physically present before you can log a working day.**

2. The Problem Being Solved

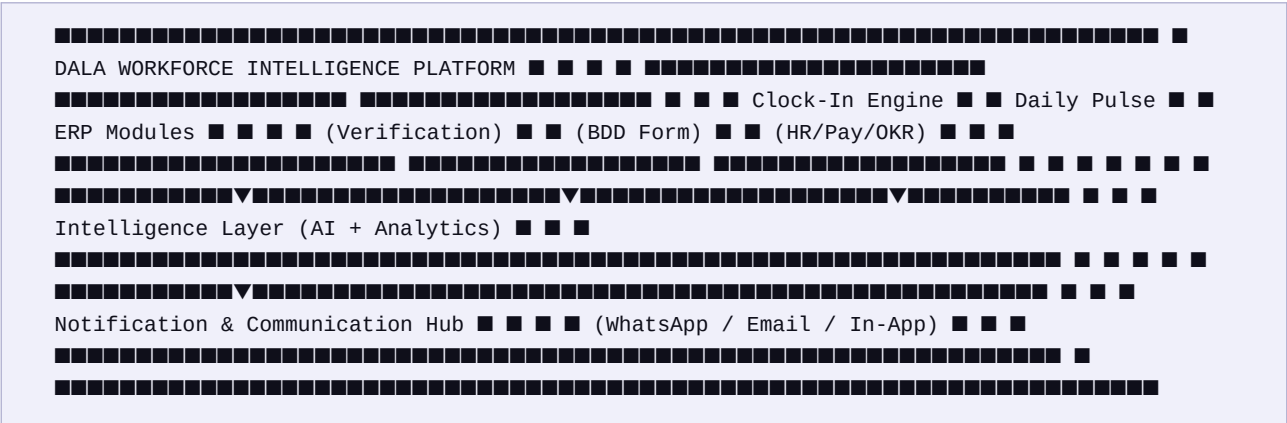
The current process uses handwritten weekly BDD sheets capturing:

- Daily priorities and accomplishments
- Blockers and support needed
- Weekly goal progress
- AI usage reflection (Saturdays)

Problems with the paper process:

- No verification the person was physically present
- Data is not searchable, reportable, or analyzable
- No link between daily output and payroll
- Easy to falsify or fill in retroactively
- Manager must manually read 10–20 forms per week
- No early warning for recurring blockers or absenteeism

3. System Architecture



Tech Stack:

- **Frontend:** React + TypeScript + Tailwind CSS (shadcn/ui)
- **Backend:** Node.js + Express
- **Database:** PostgreSQL via Prisma ORM
- **AI/ML:** Face recognition (@vladmandic/face-api), Claude API for text summarization
- **Real-time:** WebSockets (Socket.io)
- **Background Jobs:** Trigger.dev
- **Notifications:** Twilio (WhatsApp/SMS) + Firebase Cloud Messaging
- **Deployment:** Railway

4. Clock-In Verification — Phased Rollout

The verification stack is built in phases. Each phase adds a hardware layer on top of the previous one. The app logic handles all phases — you upgrade the hardware, not the software.

Phase 0 — QR Code (Current, No Hardware Required)

What it uses: A screen at each entry point + any smartphone camera.

How it works:

A cheap device (old phone, tablet, spare laptop) is mounted at each entry point and loads a single rotating QR page. The server generates a brand new code every 3 minutes. The old code is immediately invalidated.

Each QR token contains: ■■■ Entry point ID (which door) ■■■ Timestamp (when generated) ■■■ One-time token (cryptographic string) ■■■ Expiry (3 minutes)

Employee clock-in flow:

Employee arrives at door ↓ Opens Dala app → taps Clock In ↓ Camera opens → scans QR on screen ↓ Server validates token (not expired, not used before) ↓ Face scan (live photo vs. master photo) ↓ Liveness check (blink prompt – prevents photo spoofing) ↓ Daily Pulse BDD form (mandatory, cannot skip) ↓ Clock-in complete ✓

How cheating is prevented:

Attempt	Why It Fails
Screenshot QR, send to absent colleague	Token expires in 3 minutes
Use old screenshot	Token is one-time use — consumed on first scan
One person scans for others	Face recognition catches the wrong person immediately after
Scan from outside through a window	Must be within close camera range of the screen
Manipulate the app	Server validates all tokens — client cannot forge them

Hardware needed per entry point:

- Any screen: old Android phone, cheap tablet, or spare laptop
- Mounted at eye level, facing employees as they walk in
- Loads: `dala.app/entry/door-[id]` — nothing else
- Cost: ■0 if spare devices available / ■30,000–■50,000 for a new tablet

Phase 1 — NFC Tags (Next Step)

What it adds: NFC sticker on each door frame. Employee taps their phone against it before the QR scan. Confirms they touched that exact entry point.

Requirement: Employee phone must have NFC hardware.

Nigerian phone reality: Budget phones (Tecno Spark, Itel, old Androids) often do not have NFC. Therefore NFC is **optional** — employees without NFC phones fall back to QR automatically. No one is blocked.

Employee WITH NFC phone: ■■■ NFC tap → QR scan → Face scan → Liveness ✓ Employee WITHOUT NFC phone: ■■■ QR scan → Face scan → Liveness ✓

Where to buy NFC tags (Nigeria):

Option	Product	Price	Link
Jumia Nigeria	Yiliker 10pcs NTAG215 stickers	■5,053	jumia.com.ng/mlp-nfc-tags
Jumia Nigeria	50pcs NTAG216 stickers	■9,990	jumia.com.ng/mlp-nfc-tags
Jumia Nigeria	RFID White Cards 10pcs	■6,500	jumia.com.ng/mlp-nfc-tags

Option	Product	Price	Link
AliExpress	100pcs NTAG213 stickers	~₦8,000	aliexpress.com

Recommended purchase: 50pcs NTAG216 for ₦9,990 on Jumia — covers a full team and arrives in 2–3 days.

NFC chip recommendation: NTAG213 or NTAG215. NTAG213 (144 bytes) is sufficient for an employee ID and entry token. NTAG216 (888 bytes) is overkill but fine.

Phase 2 — BLE Beacons (Full Stack)

What it adds: Small Bluetooth devices mounted inside the building. The phone passively detects beacon signal — if detected at sufficient strength, the person is confirmed physically inside the building.

Why this matters: NFC and QR prove you were at the door. BLE proves you are inside the building throughout the session — passive and continuous.

Beacon signal is controllable: Radius can be tuned (5–30 metres) so it does not bleed significantly outside a solid standalone building.

BLE is NOT available on Jumia Nigeria. The "bluetooth adapters" shown on Jumia are USB dongles for computers — a completely different product. Do not buy those.

Where to buy BLE beacons (Nigeria):

Option	Product	Price	Notes
Dirigible Nigeria (Lagos)	Teltonika BTSID1	Call for price	+234 818 024 5741 / sales@dirigible.com.ng
AliExpress	NRF51822 iBeacon (BLE 4.2)	~₦8,000–₦12,000/unit	15–30 day delivery
AliExpress	Minew E7 (BLE 5.0)	~₦10,000–₦20,000/unit	Better quality
Computer Village, Ikeja	IoT/Arduino component vendors	Variable	Ask for "iBeacon BLE module"

Recommended beacons by budget:

- **Budget:** NRF51822 generic (AliExpress ~\$5) — good for testing
- **Mid-tier:** Minew E7 (AliExpress ~\$8–\$12) — better battery, BLE 5.0
- **Premium/local support:** Teltonika BTSID1 (Dirigible Nigeria)

Deployment: 1 beacon per entry zone + 1–2 inside main work area. A 3-door office needs 4–5 beacons total.

Full Verification Stack (Phase 2 Complete)

Layer	Method	Phase	Presence Proof
1	QR Code (rotating, 3-min expiry)	0 — Now	At the door

Layer	Method	Phase	Presence Proof
2	NFC Tag tap (optional fallback to QR)	1 — Soon	Exact door frame
3	BLE Beacon detection	2 — Later	Inside building
4	Face Recognition	0 — Now	Identity
5	Liveness Detection	0 — Now	Anti-spoof
6	Google Fused Location	0 — Now	Audit log only

Note on GPS / Google Maps: Raw GPS is unreliable indoors. The system uses Google's Fused Location Provider (combines GPS + visible WiFi signals + cell towers) as a **background audit tag only** — it logs approximate coordinates for records but does not block or gate the clock-in. It is most useful for remote/field workers.

5. Module 2 — Daily Pulse (Digital BDD Check-In)

This is a **mandatory form that appears immediately after a successful clock-in**. It cannot be dismissed or skipped. It directly replaces the handwritten BDD weekly sheet.

Monday Form — Weekly Brief

1. What did you accomplish last week? [Long text] 2. What is your top priority / goal this week? [Long text] 3. Any challenges you foresee this week? [Long text] 4. What support do you need from management? [Long text] 5. Today's top 1-3 priorities [3 fields] 6. Questions / Notes [Long text] 7. Previous S.E. Meeting feedback [Long text]

Tuesday – Friday Form — Daily Standup

1. Progress toward weekly goal [0-100% slider] 2. What did you complete yesterday? [Long text] 3. Today's top 1-3 priorities [3 fields] 4. Any blockers? [Long text] 5. Support / assistance needed [Long text] 6. Questions / Notes [Long text]

Saturday Form — Weekly Wrap

1. Was your weekly goal achieved? [Yes / Partial / No + reason] 2. Key wins this week [Long text] 3. What you achieved WITH AI this week [Long text] 4. What would you do differently? [Long text] 5. Top 1-3 priorities for next week [3 fields]

AI Layer on the BDD Form

- Each submission is summarized into one sentence by Claude API

Every clock-in data point has a direct financial consequence. The system does not separate attendance from pay.

BDD Data Point	Payroll Impact
Weekly goal completion rate ≥ 80%	Bonus eligibility unlocked
Consecutive BDD submissions (no skips)	Consistency bonus
Skipped BDD form after clock-in	Warning → docked from performance score
Same blocker unresolved 3+ days	Under-performance review flag
Saturday AI usage field	AI adoption incentive (if company runs programme)

Leave Type	Pay Rule
Annual Leave	Full pay
Sick Leave	Full pay (within allowance)

Leave Type	Pay Rule
Casual Leave	Full or half pay (policy-defined)
Study Leave	Partial pay
Unpaid Leave	Zero pay
Absent (no request filed)	Full day deducted + disciplinary flag

The Audit Trail Advantage

Because every action is timestamped, photo-verified, and GPS-tagged:

- No disputes about "I was there" — QR token + face + liveness proves it
- BDD timestamps prove when work planning occurred
- Full exportable log per employee per month
- Payroll disputes resolved from immutable records

Payroll Outputs

- Individual payslip PDF per employee
- Bank transfer summary export (Excel)
- Month-to-date deduction breakdown
- Employee dispute mechanism (flag a line item for manager review)

7. ERP Core Modules

7.1 HR Management

- Employee profiles: personal info, role, department, contract type
- Org chart view
- Document storage: contracts, ID, certificates (with expiry alerts)
- Onboarding digital checklist per role
- Offboarding workflow

7.2 Leave Management

- Leave balance tracker per employee per type
- Manager approval workflow (approve / reject / request clarification)
- Team calendar — see who is out on any given day
- Automatic payroll deduction on approval

7.3 Expense Management

- Submit expenses with receipt photo

- Category tagging (travel, meals, supplies, client entertainment)
- Manager approval workflow
- Approved expenses added to payroll reimbursement line
- Monthly expense summary report per employee and department

7.4 Performance Management (OKR)

- Quarterly objective setting per employee
- Key results with progress tracking
- BDD daily data feeds performance score automatically
- Manager review cycle (monthly / quarterly)
- Streak tracking: longest consecutive on-time days with completed BDD forms

7.5 Training and Certification Tracking

- Log completed training per employee
- Certification expiry alerts (30 days and 7 days before expiry)
- Required vs. completed training matrix per role
- Training completion visible in performance reviews

7.6 Shift and Schedule Management

- Assign shifts and working hours per employee
- Overtime pre-approval requests
- Shift swap requests (peer-to-peer, manager-approved)
- Late threshold configurable per shift (not hardcoded to 9:00 AM)

8. Intelligence Layer

Feature	Description
Attendance Prediction	Identifies employees likely to be absent based on historical patterns
Pattern Anomaly Alerts	"Employee X has been late every Monday for 6 weeks" → manager alert
AI Team Brief	Auto-generated Monday narrative of team priorities via Claude API
Productivity Correlation	BDD goal completion rate vs. attendance consistency graph
Mood Trend Dashboard	Team-level emotion trends over time (anonymized aggregate)
AI Adoption Tracker	Saturday BDD field aggregated into weekly team report
Blocker Detection	Same blocker logged 3+ days → escalated to manager automatically

Feature	Description
Absence Escalation	No clock-in by 10 AM with no leave on file → automatic manager alert

9. Notification and Communication Hub

Automated Alerts

Trigger	Recipient	Channel
Not clocked in by threshold time	Employee	WhatsApp / SMS
Absent, no leave request, by 10 AM	Line Manager	Email + In-App
Late 3+ consecutive days	HR	In-App alert
Same blocker 3 days running	Manager	In-App alert
Leave request submitted	Manager	Email + In-App
Leave approved or rejected	Employee	WhatsApp + In-App
Payslip ready	Employee	Email + In-App
Certification expiring in 30 days	Employee + Manager	Email

Manager Daily Digest (7:00 AM)

- Who is clocked in (count and names)
- Who is late and by how many minutes
- Who is absent and whether leave is filed
- Pending approvals: leave, expense, corrections
- Team mood summary from previous day's clock-ins

Broadcast System

- Admin sends office-wide or department-specific broadcasts
- Broadcasts appear on employee dashboard on next login
- Expiry time configurable per broadcast

10. Multi-Location and Hybrid Work

- Define multiple office locations (HQ, branch, client sites)
- Each location has its own QR entry screens and (later) beacons

- **Work From Home mode:** Face + liveness check, BDD form mandatory, Fused Location logs home coordinates
- **Field Work mode:** GPS trail logged throughout the day, pre-approved by manager before day starts
- **Client Site mode:** Manager pre-registers the site, employee clocks in at client location
- All modes visible to admin on map view

11. Data Model — Key Tables

Table	Purpose
User	Employee profiles, roles, master photo, hourly rate
AttendanceRecord	Every clock-in/out with QR token, photo, method, mood, hours
QRToken	Generated tokens with entry point, expiry, used status
BDDCheckIn	Daily pulse form responses, AI summary, goal progress
CorrectionRequest	Employee-requested fixes to attendance records
LeaveRequest	Leave applications with type, dates, approval status
ExpenseRequest	Expense claims with receipt, amount, approval status
PerformanceGoal	Quarterly OKRs per employee
Shift	Scheduled shifts and hours per employee
BroadcastMessage	Office announcements with expiry
Notification	All system notifications with read status
AuditLog	Immutable record of every system action

12. Daily User Experience

Employee

7:45 AM → Phone notification: "Time to check in" ↓ Walks to entry door ↓ Opens Dala app → Clock In → camera opens ↓ Scans rotating QR on door screen ↓ Face scan → liveness check (blink) ↓ Daily Pulse BDD form (2-3 minutes) ↓ "Good morning [Name]. Top priority today: [from BDD form]" 5:30 PM → Notification: "Don't forget to clock out" ↓ Scans QR at exit → face scan ↓ "You worked 8.5 hours. 0.5 hours overtime logged."

Manager

7:00 AM → Email digest: 18 in ✓, 2 late, 1 absent (no leave on file) ↓ Dashboard: 1 leave request, 2 expense claims pending ↓ 8:30 AM → AI Team Brief: "This week the team is focused on..." ↓ One-click approvals for routine items

13. Security and Compliance

- JWT authentication with refresh tokens and MFA
- Role-based access control: Admin / Manager / Employee
- QR tokens are single-use, server-generated, cryptographically signed
- All photos and biometric data encrypted at rest
- Audit log for every action (who, what, when, from where)
- NDPR-aligned data handling (Nigeria Data Protection Regulation)

14. Hardware Summary

Phase 0 — What You Need Right Now

Item	Purpose	Cost
1 screen per entry point (old phone, tablet, or laptop)	Display rotating QR	■0 if spare devices available
Internet connection for the screen	Load QR page	Already available (company laptops are online)

Total hardware cost for Phase 0: ■0

Phase 1 — NFC Tags (Next Purchase)

Item	Where	Cost
50pcs NTAG216 sticker tags	Jumia Nigeria	■9,990

Delivery: 2–3 days (Jumia Express).

Note: NFC is optional in the app. Employees without NFC phones automatically fall back to QR. No one is blocked.

Phase 2 — BLE Beacons

Important: BLE beacons are NOT available on Jumia Nigeria. The "Bluetooth adapters" on Jumia are USB dongles for computers — a completely different product.

Option	Where	Cost per Unit	Lead Time
NRF51822 iBeacon	AliExpress	~₦8,000–₦12,000	15–30 days
Minew E7 (BLE 5.0)	AliExpress	~₦10,000–₦20,000	15–30 days
Teltonika BTSID1	Dirigible Nigeria, Lagos	Call for price	Days

Dirigible Nigeria contact: +234 818 024 5741 / sales@dirigible.com.ng

Quantity needed: 1 per entry zone + 1–2 inside the main work area. A 3-door office needs 4–5 beacons.

Full Budget Estimate (20–50 person office)

Phase	Item	Cost
Phase 0	Entry screens (spare devices)	₦0
Phase 1	50 NFC tags (Jumia)	₦9,990
Phase 2	4–5 BLE beacons (AliExpress)	₦40,000–₦100,000
Total		₦50,000–₦110,000

15. Build Phases

Phase	What Gets Built	Priority
Phase 1	QR clock-in engine + rotating token system + entry screens	Critical
Phase 2	BDD Daily Pulse form (mandatory post-clock-in gate)	Critical
Phase 3	JWT auth upgrade + MFA + real-time WebSockets	High
Phase 4	Payroll engine (deductions, overtime, bonuses, payslip PDF)	High
Phase 5	Leave and expense management with approval workflows	High
Phase 6	NFC tap layer (optional, falls back to QR)	Medium
Phase 7	Performance / OKR module + BDD feeding performance scores	Medium
Phase 8	AI summarization (Claude API) — Team Brief + blocker detection	Medium
Phase 9	Notification hub (WhatsApp / SMS + email digests)	Medium

Phase	What Gets Built	Priority
Phase 10	BLE beacon layer + hybrid/remote work modes	Low
Phase 11	Predictive ML + advanced analytics + AI adoption tracking	Low

16. Key Design Principles

- 1. Verification before entry** — no one logs attendance without proving presence
- 2. The BDD form is not optional** — it is gated, not a courtesy
- 3. Data drives pay** — every attendance point has a direct payroll consequence
- 4. Hardware phases, software stays stable** — QR today, NFC tomorrow, BLE later; the app handles all without rewrites
- 5. No employee is blocked by their phone model** — every layer has a fallback
- 6. Managers see signals, not noise** — AI surfaces patterns so humans decide
- 7. Audit everything** — every action is logged, timestamped, and non-repudiable

Dala Workforce Intelligence Platform — Internal System Design Document
Version 2.0 — Updated March 30, 2026 following planning session
Incorporates: QR-first verification, phased NFC/BLE rollout, payroll integration, BDD digital form, ERP modules, Nigerian hardware sourcing